The FFGFT Hawking Mechanism

KMS temperature, winding transition, $\mathbb{Z}_3$ triality selection, unitarity — the complete mechanism from FFGFT-native building blocks

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Abstract

Doc. 313 (Ch. G) established the thermodynamic frame of Hawking radiation in FFGFT: a single KMS rule tempers Unruh, Gibbons–Hawking and black holes jointly **[K]**, the membrane picture receives its thermometer **[S]**, and the family ladder reaches the cosmos at $r_S = R_H/2$ **[S]**. What remained open was the *microscopic emission mechanism*: which modes escape, what the quantum carries, why massive quanta fall back. This document works it out completely from FFGFT-native building blocks:

1. **Temperature**: The KMS rule $T = \hbar/(k_B\tau)$ with membrane period $\tau = 8\pi GM/c^3$ follows from $\mathfrak{T} \cdot m = 1$ (clock compression at the membrane, Doc. 313 G.2). **[K]**
2. **Quantum**: The Hawking quantum is the bit of scale $4\pi r_S$: $k_B T_H = \hbar c/(4\pi r_S)$ exactly — the system-dependent bit energy $E_{\text{bit}} = \hbar c/L$ (Doc. 257) evaluated at the horizon scale. No universal bit value is needed anywhere. **[K]**
3. **Selection**: Only triality-0 modes ($T_R = 0$) can be emitted. Confinement (Doc. 321) and Hawking selection are *one* principle; absorption of the partner quantum is the $\mathbb{Z}_3$ projector orthogonality $P_j P_k = 0$. **[B]**
4. **Carrier**: The massless carrier is the $n_4 = 0$ torus mode (spatial winding without mass-circle winding). Massive modes ($n_4 \neq 0$) have finite $\mathfrak{T}$ and fall back. **[K]**
5. **Information**: The emitted quantum carries the sector pair $(k, -k)$ as an orthogonally readable quantum number ($\log_2 3$ bits) — plus the thermal 1-nat area bookkeeping. Fine-grained entropy remains constant (unitarity, Doc. 322). No information paradox. **[K]**

Two verification scripts, all assertions passed: `325_hawking_ffgft_mechanismus.py` (thermodynamic chain), `325_hawking_z3_selektion.py` ($\mathbb{Z}_3$ algebra).

.1 Starting point: what Doc. 313 left open

Doc. 313 Ch. G established three results: (a) the single KMS rule $T = \hbar/(k_B\tau)$ for Unruh, Gibbons–Hawking and black holes **[K]**; (b) the membrane thermometer from $Ŧ \cdot m = 1$ **[S]**; (c) the family ladder up to the cosmos ($r_S = R_H/2$ exactly) **[S]**. Likewise the limit: evaporation is 56 (stellar) resp. 83 (supermassive) orders of magnitude too slow for the time cycle — Hawking sharpens the coarse-grained entropy question D(ii), it does not solve it **[K]**.

What was missing was the *microstructure* of emission:

- Which modes of the $T^4/\mathbb{Z}_3$ spectrum can leave the horizon — and which principle selects them?
- What is the emitted quantum, and how is its energy related to the information unit of the framework (Doc. 257)?
- Why do massive quanta fall back — as a consequence of the geometry, not as an extra assumption?
- How is information conservation realized microscopically?

These four questions are answered here — each from an already established FFGFT building block: $Ŧ \cdot m = 1$ (Docs. 077, 180), system-dependent bit energy (Docs. 257, 302, A271–A273), $\mathbb{Z}_3$ triality (Doc. 321), unitarity of the spectral operator (Doc. 322).

.2 Building block 1: the KMS temperature from $Ŧ \cdot m = 1$

Wherever the local time structure is periodic with period τ, an observer reads off temperature:

$$T = \frac{\hbar}{k_B \tau} \tag{1}$$

System	Period τ	$T = \hbar/(k_B\tau)$
Cosmos (compact time)	$2\pi/H_0 = 2.9 \times 10^{18}$ s	2.66×10^{-30} K
Black hole ($M_\odot$)	$8\pi GM/c^3 = 1.24 \times 10^{-4}$ s	6.17×10^{-8} K
Unruh ($a = g$)	$2\pi c/a = 1.9 \times 10^8$ s	3.98×10^{-20} K

The membrane period $\tau = 8\pi GM/c^3$ is not an extra assumption but follows from time-mass duality: mass compresses the time structure ($Ŧ \cdot m = 1$ — the closer to the membrane, the slower the clocks), and at the membrane the local time structure becomes periodic with exactly this period (Doc. 313 G.2). **[K]**

The temperature is thus not the result of a Bogoliubov computation on a curved background, but a read-off rule of the clock map — the same rule for all three systems.

.3 Building block 2: the Hawking quantum as a system-dependent bit

The central new result of this document. The system-dependent bit energy of FFGFT (Doc. 257) reads $E_{\text{bit}}(L) = \hbar c/L$. Evaluated at the horizon scale:

Core result [K]

$$k_B T_{\mathrm{H}} = \frac{\hbar c}{4\pi r_{\mathrm{S}}} = E_{\text{bit}}(L = 4\pi r_{\mathrm{S}}) \tag{2}$$

The thermal Hawking quantum *is* the bit of scale $L = 4\pi r_S$ (twice the horizon circumference). Numerically exact (script 1): $\hbar c/(k_B T_H) = 4\pi r_S$ to machine precision.

The system dependence of bit values (Docs. 257, 302, A271–A273) is therefore no obstacle to Hawking physics but *its explanation*: the Hawking temperature is the bit temperature of the horizon scale. A universal bit value — say, a fixed bit mass from a Landauer evaluation at the Planck temperature — is needed nowhere; every system carries its own bit energy, and that of the black hole at its horizon is exactly $k_B T_H$.

Area bookkeeping

Emission of a quantum of energy $k_B T_H$ carries, by Clausius, exactly $\mathrm{d}S = E/(T \cdot k_B) = 1$ nat, hence

$$\mathrm{d}A = -4\ell_P^2 \quad \text{per nat} \tag{3}$$

(script 1). The area loss per emission thus follows from Bekenstein + Clausius alone — no further geometric assumptions. **[B]**

.4 Building block 3: $\mathbb{Z}_3$ triality selection

The algebraic core of the mechanism. The horizon carries the $\mathbb{Z}_3$ orbifold structure of $T^4/\mathbb{Z}_3$ (Docs. 321, 322). The three $\mathbb{Z}_3$ projectors

$$P_k = \frac{1}{3}\sum_{j=0}^{2} \omega^{-jk}\,\tau^j, \qquad \omega = e^{2\pi i/3} \tag{4}$$

are complete, idempotent and orthogonal (script 2):

$$P_0 + P_1 + P_2 = \mathbb{1}, \qquad P_j P_k = \delta_{jk} P_j. \tag{5}$$

Absorption [B]

The infalling partner quantum erases the horizon voxel exactly when it lies in the orthogonal $\mathbb{Z}_3$ sector:

$$P_j\,(P_k\,\psi) = 0 \quad (j \neq k) \tag{6}$$

— exact, no tunnelling (script 2: $\|P_2(P_1\psi)\| < 10^{-15}$). Pair creation at the membrane separates a $(k, -k)$ sector pair; the inward-falling partner is absorbed by projector orthogonality, the outward-running one escapes provided it satisfies the triality condition.

Who escapes: $T_R = 0$

The triality of a mode is its $\mathbb{Z}_3$ charge k. Emittable (asymptotically free) are only combinations with $T_R = \sum k_i \equiv 0 \pmod{3}$:

Combination	T_R	emittable
Single mode $k = 0$ (photon analogue)	0	yes
Single mode $k = 1$ (quark)	1	no
Bilinear $(1, 2)$ (meson)	0	yes
Bilinear $(1, 1)$	2	no
Trilinear $(1, 1, 1)$ (baryon)	0	yes

Confinement = Hawking selection [B]

Doc. 321 identified confinement as triality selection $T_R = 0$. The same principle acts at the horizon: only colourless objects escape. Confinement and Hawking selection are *one* principle of the $\mathbb{Z}_3$ structure — not two. The $\mathbb{Z}_3$ charge is exactly conserved in every emission (script 2).

.5 Building block 4: the massless carrier

On $T^4 = T^3_{\text{space}} \times S^1_m$ the mass of a mode is carried by the winding n_4 on the mass circle ($Ŧ \cdot m = 1$). The emittable carrier is the mode with

$$n_4 = 0 : \qquad \text{spatial winding without mass winding.} \tag{7}$$

A massive quantum ($n_4 \neq 0$) has finite Ŧ; in the compressed clock map at the membrane (Doc. 313 G.2) it falls back — as a consequence of the geometry, not as an extra assumption. For $|n_i| \leq 1$ there are $3^3 - 1 = 26$ massless neighbouring modes against 54 massive ones (script 2). **[K]**

The Hawking spectrum thus consists of $n_4 = 0$ modes with $T_R = 0$: massless, colourless, thermal at T_H — the photon-like degrees of freedom of the torus.

.6 Building block 5: information conservation

What the quantum carries

The emitted $T_R = 0$ quantum carries the *sector pair* $(k, -k)$ as an internal quantum number. The three possible pair states are exactly orthogonal (Gram matrix $= \mathbb{1}_3$, script 2) — which sector was withdrawn is readable from the quantum: $\log_2 3 \approx 1.585$ bits of sector information per quantum, in addition to the thermal 1-nat area bookkeeping.

Fine-grained constancy

Doc. 313 G.5 phrased it as "two languages, one statement": locally, the radiation is correlated with the fractal membrane structure ($S_\text{BH}(M_\odot) = 1.05 \times 10^{77}\, k_B$ of correlations — nothing is lost); globally, fine-grained entropy is constant under unitary evolution (unitarity of the spectral operator, Doc. 322 **[K]**). No information paradox: the membrane is a clock-map object, not a causal abyss.

.7 The FFGFT correction and its limits

Doc. 313 gives the fractal correction of the Hawking power:

$$P = P_\text{std}\,\big(1 - \xi \ln(M/m_\text{P})\big), \tag{8}$$

numerically 0.60 % (PBH 10^{12} kg) to 1.44 % (SMBH $10^9\, M_\odot$) — script 1. It does not change the evaporation findings: only primordial holes below

$$M_* = \left(\frac{\tau_c\, \hbar\, c^4}{5120\, \pi G^2}\right)^{1/3} = 3.3 \times 10^{11}\ \text{kg} \tag{9}$$

close via Hawking within one time cycle $\tau_c = 91.9$ Gyr. The Sun is 56, an SMBH 83 orders of magnitude too slow (script 1) — Hawking sharpens the coarse-grained entropy question D(ii), it does not solve it (Doc. 313 G.4). **[K]**

The family ladder runs seamlessly up to the cosmos: $T_\mathrm{H} = T_\mathrm{GH}$ requires $M = c^3/(4GH_0) = 4.66 \times 10^{52}$ kg with $r_\mathrm{S} = R_H/2$ exactly (script 1). **[K]**

Placement of the bit result

$k_B T_\mathrm{H} = E_\mathrm{bit}(4\pi r_\mathrm{S})$ shows: Hawking physics can be formulated entirely with the system-dependent bit energy. Any construction that instead postulates a universal bit value (say, a fixed bit mass from $k_B T \ln 2$ at a privileged temperature) adds a parametrization that the temperature physics does not require — consistent with A271–A273: Landauer's bound is a statement about carrier operations, not about abstract information.

.8 Status overview

Element	Status
KMS rule tempers Unruh/GH/BH jointly	**[K]** (Doc. 313)
Membrane period from $\mathrm{T} \cdot m = 1$	**[K]**
$k_B T_\mathrm{H} = E_\mathrm{bit}(4\pi r_\mathrm{S})$ — quantum = horizon bit	**[K]** (new)
Area quantum $-4\ell_\mathrm{P}^2$/nat from Bekenstein+Clausius	**[B]** (new)
$\mathbb{Z}_3$ absorption $P_j P_k = 0$	**[B]** (new)
$T_R = 0$ selection; confinement = Hawking selection	**[B]** (new)
Carrier = n_4-free torus mode	**[K]** (new)
Sector info $\log_2 3$ bits orthogonally readable	**[B]** (new)
Information conservation (membrane + unitarity)	**[K]** (313/322)
FFGFT power correction 0.6–1.4 %	**[K]** (313)
Greybody factors, mode-spectrum detail	**[S]** (open)
Back-reaction on orbifold fixed points	**[S]** (open)

The Hawking mechanism in FFGFT thus stands at **[K]**/**[B]**; open remain only the greybody factors (standard QFT computation on the FFGFT background) and the back-reaction of the emission on the orbifold fixed-point structure.

Verification scripts

- `python/Dok325_Skripte/325_hawking_ffgft_mechanismus.py` — thermodynamic chain: KMS, membrane thermometer, horizon bit, area bookkeeping, correction, evaporation, family ladder, information side.
- `python/Dok325_Skripte/325_hawking_z3_selektion.py` — $\mathbb{Z}_3$ algebra: projectors, absorption, triality selection, charge conservation, sector information, masslessness.

References

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